General Assumptions and Improvements / Open Points

Embedded Temp System

Content

01 General Assumptions

02 Improvements / Open Points

01 General Assumptions

- No requirements for scheduling sw functions, therefore all are executed/call in idle task/main loop
- Due to hard Adc sampling requirement of 100us with low jitter, in function “Init_Read_Adc”, an internal uC Timertrigger for Adc and storage with DMA (direct memory access) is initialized
- EEPROM HW-Rev is part of serial number and can be red out of EEPROM (therefore EEPROM-HW is realized as separate SW (mocked-HW) module “read_mockedEeprom_dev”
- Temp Range of “WARNING” and “CRITICAL” are overlapping, so there are situations where both LEDs ON
- AdcSens selection is realized via application variable in external application file (so changing Sensor without rebuilding SW is possible)
- Adc Input Voltages are defined between 0V and 5V, only 11 Values are possible for test purpose
- Temperature Range is defined from -25°C to 125°C
- Error handling and limiter to (max/min) values only basically realized or skipped

02 Improvements / Open Points

- Error handling and limiter (max/min) improvement
- Clean Up (not used headers / functions/variables)
- Replace data transfer via transfer_struct with a proper message concept
- Replace floor with round (do to some tool issue in ide)
- Introduce basic OS with scheduling options for functions
- Optimize functions and algorithms to short and more efficient code